

HIGH-PRECISION ALIGNMENT OF AN UPGRADED SOFT X-RAY POLARIMETER AT DIAMOND LIGHT SOURCE

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Abstract

Knowledge of X-ray beam polarisation on a synchrotron beamline is essential, not only for characterising the undulator performance, but also for precise analysis of dichroic and chiral experiments. The upgraded high-precision soft X-ray polarimeter at Diamond Light Source features multiple retarder adjusters to allow precise concentricity and angular alignment to the analyser. An offline alignment procedure has been developed, achieving 69 μm horizontal and 17 μm vertical concentricity alignment, as well as 4 μrad yaw and 9 μrad pitch alignment. Compared to the original version of the instrument, the vertical concentricity alignment improved by 14 $\times$ and the pitch alignment improved by 18 $\times$. The procedure uses a laser diode to mimic the X-ray beam and a hexapod to align the analyser. Concentricity alignment relies on monitoring the intensity as the laser beam is cropped by plane mirrors on the retarder and analyser stages. Angular alignment is achieved by measuring the retarder and analyser rotation vectors using an autocollimator. The improved alignment allows the polarimeter to meet the stringent requirements for complete polarisation measurement above 1 keV.

INTRODUCTION

A polarimeter uses a retarder to introduce phase retardation and an analyser to transmit only one linearly polarised component [1]. The independent rotation of these polarising elements along the beam axis allows polarisation analysis at synchrotron sources to characterise the performance of undulators capable of providing linear and circular polarisations in the soft X-ray region [2]. Highly polarised radiations are utilised in beamlines to perform experiments including X-ray magnetic linear/circular dichroism [3, 4] and X-ray magnetic scattering [5].

Modern soft X-ray polarimeters use multilayer and crystal polarising elements as transmission phase retarders and reflection analysers [6]. At energy above 1 keV, the retarder must be operated at high grazing angle to maximise phase shift [7]. Assuming a sample size of 5 mm and a grazing angle of 10°, the effective sample size is only 870 μm . In addition, the Bragg resonance bandwidth of the polarising elements is very narrow at higher energies [8]. These factors impose a tight tolerance on the concentric and angular alignment between the polarising samples.

The original soft X-ray polarimeter at Diamond Light Source featured a conventional design that lacked alignment adjustability. It had a vertical concentricity offset of 350 μm and a pitch offset of 160 μrad because of mechanical imperfections [9]. These misalignments introduced a modulation of the angle of incidence with a period 2π leading to considerable measurement error when characterising the degree of polarisation of undulators at higher energies. To achieve complete polarisation analysis above 1 keV, the instrument went through a complete redesign to incorporate multiple adjusters on the retarder for concentricity and angular alignments.

Figure 1a shows a render of the upgraded polarimeter. The retarder rotation axis rests on top of a base plate that can be translated as shown in Fig. 1b. Adjusting adjacent pairs of adjustment screws by different amounts also allows for coarse angle alignment. Fine angle alignment can be performed using adjuster screws on top of the retarder rotation axis as shown in Fig. 1c. The rest of this paper introduces an offline alignment procedure developed to improve vertical concentricity alignment 14 times and pitch alignment 18 times compared to the original version of the instrument.

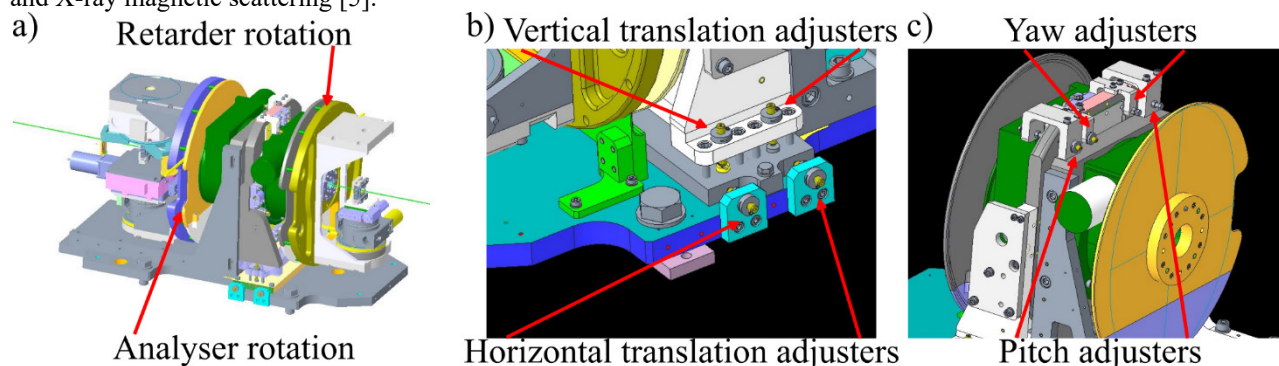


Figure 1: a) The upgraded soft X-ray polarimeter at Diamond. Manual adjusters incorporated on the retarder axis for b) horizontal and vertical translation, and c) yaw and pitch adjustment.

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EXPERIMENTAL SETUP

The alignment setup is shown in Fig. 2. It used a laser diode to shine a visible beam through the polarimeter from the retarder end to mimic the X-ray beam. An autocollimator was setup on the analyser end with the measurement beam also shining through the polarimeter. Two plane mirrors were mounted in position of the analyser and retarder samples. Rotation of the detector arm gave access to another three alignment tools mounted on the detector arm. A plane mirror facing the autocollimator, a photodetector without aperture, and a photodetector with slit aperture.

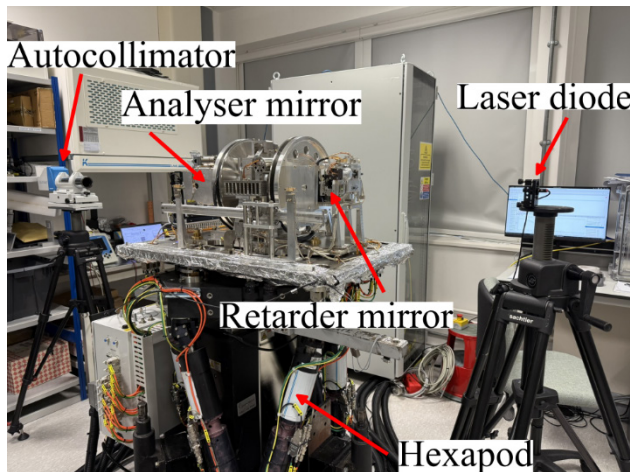


Figure 2: Photograph of the setup used to align the polarimeter in a peripheral lab before beamline operation.

OFFLINE ALIGNMENT PROCEDURE

Analysers to Laser Beam Alignment

Prior to relative alignment between the retarder and analyser, it is essential to ensure the fixed analyser end is aligned to the laser beam by adjusting the hexapod the polarimeter sits on. The centre of rotation alignment was performed by following the principle illustrated in Fig. 3a. In this experiment, the photodiode intensity was recorded as a function of mirror position at analyser rotation angles 0° , 90° , 180° and 270° . Each dataset was analysed by performing a differentiation followed by a Gaussian fit to identify the laser position where the fit peaks. Half the difference in peak position at 0° and 180° indicates the separation from the horizontal centre of rotation to the laser beam. Similarly, half the difference in peak position at 90° and 270° indicates the separation from the vertical centre of rotation to the laser beam. Figures 3b and 3c show translating the hexapod by these separation amounts aligned the analyser centre of rotation and the laser beam to $41 \mu\text{m}$ horizontally and $28 \mu\text{m}$ vertically.

The analyser rotation vector alignment was performed by following the principle illustrated in Fig. 4a. Without changing photodetector angle, the beam intensity was recorded as a function of mirror tilt at analyser rotation angles 0° , 90° , 180° , and 270° . If the rotation vector is aligned to the transmission direction of the laser beam, the four datasets should overlap when plotted together. Figures 4b and 4c show adjustment of the hexapod until the traces

overlapped by eye. In this alignment step, it is critical for the mirror to tilt about the horizontal centre of rotation identified in the previous step. This was ensured by tilting the mirror 90° and scanning it across the laser beam while monitoring the transmitted beam intensity using the photodiode without aperture. Applying a differentiation and a Gaussian fit identified the optimum axis of mirror tilt at the peak position.

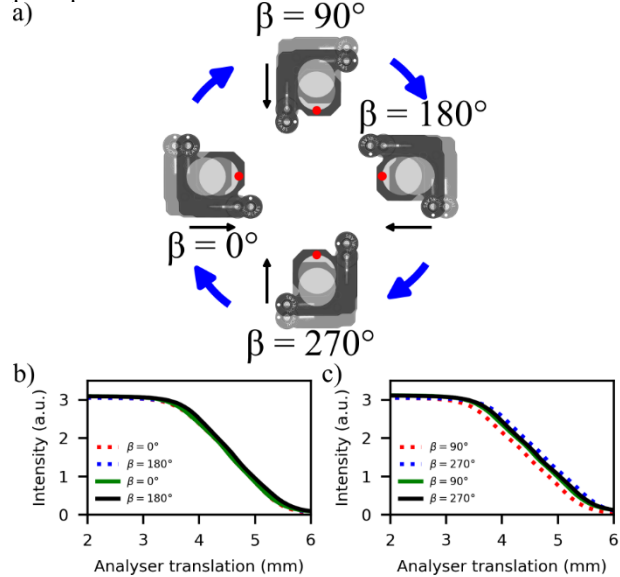


Figure 3: a) Translational alignment principle by translating the analyser mirror to crop the laser beam while monitoring the transmitted beam intensity at four analyser rotation angles (β). Note, the laser position indicated by the red dots in the cartoon is constant. Analyser centre of rotation to laser beam alignment in the b) horizontal and c) vertical directions. The dotted and solid lines show data collected before and after hexapod translation adjustment respectively. In the vertical direction, the reduction in trace separation is a clear indicator of alignment improvement.

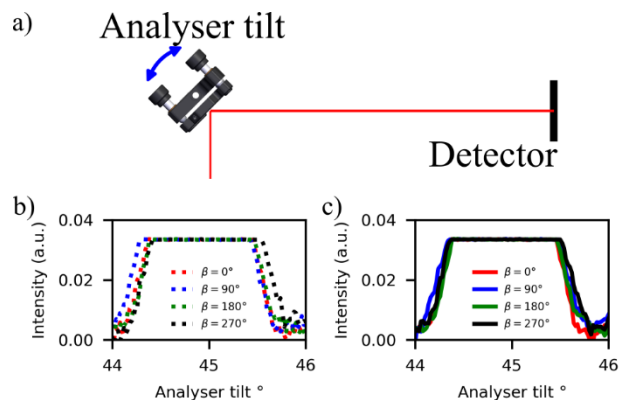


Figure 4: a) Angular alignment principle by scanning the analyser mirror tilt about 45° to deflect the laser beam into the photodetector with slit aperture at four analyser rotation angles (β). Analyser rotation vector to laser beam alignment showing b) trace offset before and c) trace overlap after hexapod angle adjustment.

Retarder to Analyser Alignment

Concentricity alignment requires a repeat of the experiment and analysis described in the first step of the previous section on the retarder. The horizontal concentricity offset is the difference between the horizontal centre of rotation to laser beam separation for the retarder and the analyser. Similarly, the vertical concentricity offset is the difference between the vertical centre of rotation to laser beam separation for the retarder and the analyser. Since the retarder translation adjustment is coupled with coarse angle adjustment, multiple translation and angle adjustment iterations are needed. Manual retarder translation adjustment reduced concentricity offset to 69 μm horizontally as shown by Fig. 5a and Fig. 3b, and 17 μm vertically as shown by Fig. 5b and Fig. 3c.

During horizontal concentricity alignment, the retarder mirror was moved to the average of Gaussian peak positions measured at 0° and 180° . This gave an indicator of the translation direction as the retarder mirror should exactly half cut the laser beam when fully aligned as shown in Fig. 5c. The same method was used to assist manual adjustment in vertical concentricity alignment.

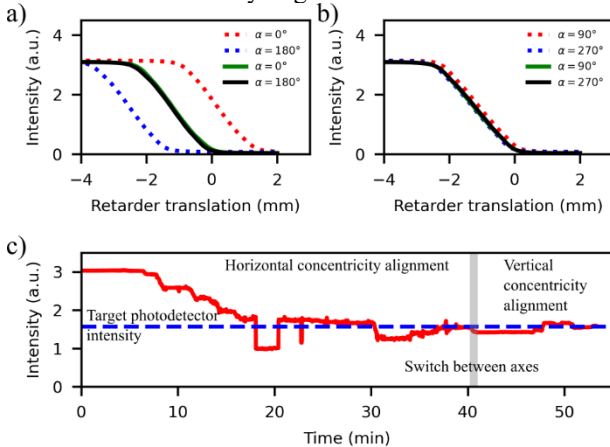


Figure 5: Retarder to analyser concentricity alignment in the a) horizontal and b) vertical directions showing data collected at four retarder rotation angles (α). The dotted and solid lines show data collected before and after retarder translation respectively. In the horizontal direction, the reduction in trace separation is a clear indicator of alignment improvement. c) Typical photodetector detector intensity timeseries during an iteration of concentricity adjustment. Manual adjustment was made until the transmitted beam intensity reached half the maximum intensity measured in the retarder concentricity alignment scans.

Relative angular alignment was done using the autocollimator to measure the analyser and retarder rotation vectors. The analyser rotation vector was measured by projecting the autocollimator measurement beam at the plane mirror mounted on the detector arm. In this experiment, the analyser rotation angle was scanned bidirectionally between 0° and 350° with a step size of 10° . A set of autocollimator measurements were recorded at each step. The data was processed by fitting a circle to find the rotation vector at the origin. Repeating this experiment and analysis on the

retarder gave the retarder rotation vector. The angular offset is the difference between these rotation vectors. Since the retarder mirror is not mounted on the pitch and yaw rotation axes, angular adjustment can change the concentricity offsets measured, resulting in the need of multiple translation and angle alignment iterations. Figure 6 shows the relative angular alignment achieved by manually adjusting retarder angle with live feedback from the autocollimator.

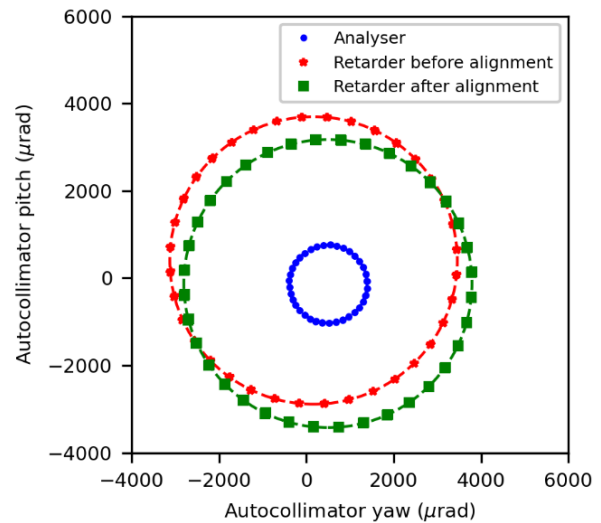


Figure 6: Retarder to analyser angular alignment. In each dataset, the origin indicates the rotation vector, and the radius indicates the mirror tilt. Relative to the analyser rotation vector, the angular offset reduced from 330 μrad in yaw and 541 μrad in pitch to 4 μrad in yaw and 9 μrad in pitch.

CONCLUSION

This paper introduced the upgraded soft X-ray polarimeter at Diamond Light Source. Modifications of the instrument incorporated mechanical alignment adjustability on the retarder for relative alignment to the analyser. To complement the hardware upgrade, a high-precision alignment procedure was developed that uses the laser beam emitted by a laser diode to mimic the X-ray beam in a peripheral lab. Cropping the laser beam using plane mirrors mounted in position of polarisation sample improved vertical concentricity alignment 14 times compared with the original version of the instrument. Measuring the rotation vector of the two axes using an autocollimator improved the pitch alignment 18 times compared to the original version of the instrument. This sets the foundation to achieving complete polarisation analysis above 1 keV.

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